

Q:- An agricultural sprinkler distributes water in a circular pattern of radius 100ft. It supplies water to a depth of e^{-r} feet per hr at a distance of r ft from the sprinkler.

(a) If $0 < R \leq 100$, what is total amount of water supplied per hr to the region inside the circle of radius R centered at sprinkler?

Sol
$$2\pi \int_0^R \int_0^{2\pi} e^{-r} r dr d\alpha$$

$$\left(\int_0^{2\pi} d\alpha \right) \left(\int_0^R e^{-r} r dr \right)$$

$$2\pi \int_0^R e^{-r} r dr$$

Let $u = r$, $v = e^{-r}$

$$2\pi \left[r \int e^{-r} dr - \left[\int \left(\frac{d}{dr} \cdot r \right) \left(\int e^{-r} dr \right) \right] dr \right]_0^R$$

$$2\pi \left[r(-e^{-r}) - \left[\int 1 \cdot (-e^{-r}) \right] dr \right]_0^R$$

$$2\pi \left[r(-e^{-r}) + (-e^{-r}) \right]_0^R$$

$$2\pi \left[-e^{-R} (1+R) + 1 \right]$$

$$2\pi \left[1 - \frac{R+1}{e^R} \right]$$

(a) Find out total amount of water supplied per hr to region inside the circle of radius R centered at sprinkler. Also sketch the region.



$$0 \leq r \leq R$$

$$0 \leq \alpha \leq 2\pi$$

(b) If $0 < R \leq 100$, find the total amount of water supplied.

If $R = 100$

$$2\pi \left[1 - \frac{100+1}{e^{100}} \right] \Rightarrow -2.02\pi e^{-100} + 0.2\pi$$

Q:- The population density of a city is approximated by the model
 $f(x, y) = 4000 e^{-0.01(x^2 + y^2)}$, $x^2 + y^2 \leq 121$
 where x & y are measured in miles. Integrate the density ftn
 over specified region to approximate population of city?

Sol

$$x^2 = x^2 + y^2$$

$$x^2 \leq 121$$

$$x \leq 11$$

$$0 \leq r \leq 11$$

$$0 \leq \theta \leq 2\pi$$

$$\text{Let } x = r \cos \theta, y = r \sin \theta$$

$$\int_0^{2\pi} \int_0^{11} 4000 e^{-0.01 r^2} r dr d\theta$$

$$\int_0^{2\pi} \int_0^{11} 4000 e^{-\frac{r^2}{100}} r dr d\theta$$

$$\text{Let } u = -\frac{r^2}{100}$$

$$\text{When } r=0 \quad u=0$$

$$r=11 \quad u = \frac{121}{100} = 1.21$$

$$\frac{du}{dr} = -\frac{r}{50}$$

$$-50 du = r dr$$

$$\int_0^{2\pi} \int_0^{11} (-50 e^u du) 4000$$

$$4000 \int_0^{2\pi} [-50 e^u]_0^{1.21} d\theta$$

$$4000 \int_0^{2\pi} -50 e^{-\frac{r^2}{100}} \Big|_0^{11} d\theta$$

$$4000 \int_0^{2\pi} (-50 e^{-\frac{121}{100}} + 50 e^0) d\theta$$

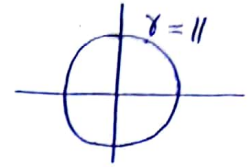
$$4000 \int_0^{2\pi} (-50 e^{-1.21} + 50) d\theta$$

$$4000 (-50 e^{-1.21} + 50) \theta \Big|_0^{2\pi}$$

$$[50 (1 - e^{-1.21})] (2\pi) 4000$$

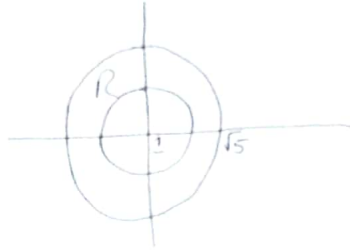
$$400000 \pi (1 - e^{-1.21})$$

$$881911$$



Q:- Let R be the annular region lying b/w two circles $x^2 + y^2 = 1$ & $x^2 + y^2 = 5$. Sketch the region, & evaluate the integral $\iint_R (x^2 + y^2) dA$

Sol



$$x^2 + y^2 = 1^2$$

$$r^2 = 1$$

$$\boxed{r = 1}$$

$$x^2 + y^2 = 5$$

$$r^2 = 5$$

$$\boxed{r = \sqrt{5}}$$

$$1 \leq r \leq \sqrt{5}$$

$$0 \leq \theta \leq 2\pi$$

$$\text{Let } x = r \cos \theta, y = r \sin \theta$$

$$\int_0^{2\pi} \int_1^{\sqrt{5}} (r^2 \cos^2 \theta + r^2 \sin^2 \theta) r dr d\theta$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\int_1^{\sqrt{5}} r^2 \left[\int_0^{2\pi} \left(\frac{r}{2} + \frac{r \cos 2\theta}{2} + \sin^2 \theta \right) d\theta \right] dr$$

$$\int_1^{\sqrt{5}} r^2 \left[\frac{r\theta}{2} + \frac{r \sin 2\theta}{4} - \cos \theta \right]_0^{2\pi} dr$$

$$\int_1^{\sqrt{5}} r^2 (\pi r) dr$$

$$\frac{\pi r^4}{4} \Big|_1^{\sqrt{5}} \Rightarrow 6\pi$$

Evaluate $\int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \sqrt{x^2+y^2} dy dx$ using Polar Coordinates. Sketch the region R of integration, specify the converted limits & the evaluate it.

$$y = \sqrt{4-x^2}$$

$$y^2 = 4-x^2$$

$$y^2+x^2=4$$

$$x^2=4$$

$$\boxed{x=2}$$

$$0 \leq r \leq 2, \quad \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$$

$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \int_0^2 r \cdot r dr d\theta$$

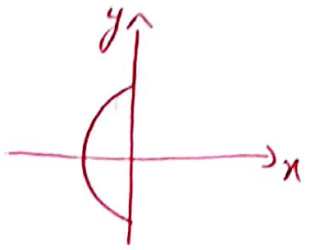
$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \left. \frac{r^3}{3} \right|_0^2 d\theta$$

$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{8}{3} d\theta$$

$$\left. \frac{8}{3} \theta \right|_{\frac{\pi}{2}}^{\frac{3\pi}{2}}$$

$$\frac{8}{3} \left(\frac{3\pi}{2} - \frac{\pi}{2} \right)$$

$$\frac{8\pi}{3}$$



Q :- Let R be the annular region lying b/w two circles $x^2+y^2=1$ & $x^2+y^2=5$. Sketch the region, specify the limits & evaluate the integral $\iint_R (x^2+y^2) dA$

Sol

$$x^2+y^2=1^2$$

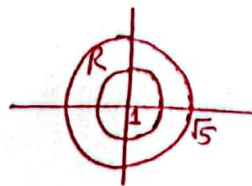
$$r^2=1$$

$$\boxed{r=1}$$

$$x^2+y^2=5$$

$$r^2=5$$

$$\boxed{r=\sqrt{5}}$$



$$1 \leq r \leq \sqrt{5}$$

$$0 \leq \theta \leq 2\pi$$

$$\text{let } x = r \cos \theta, \quad y = r \sin \theta$$

$$\int_0^{2\pi} \int_1^{\sqrt{5}} r^2 \cdot r \, dr \, d\theta$$

$$\int_0^{2\pi} \int_1^{\sqrt{5}} r^3 \, dr \, d\theta$$

$$\int_0^{2\pi} \left. \frac{r^4}{4} \right|_1^{\sqrt{5}} d\theta$$

$$\int_0^{2\pi} \left[\frac{25}{4} - \frac{1}{4} \right] d\theta$$

$$\int_0^{2\pi} 6 \, d\theta$$

$$6(2\pi)$$

$$12\pi$$